AIN SHAMS UNIVERSITY

FACULTY OF ENGINEERING

Mechatronics Engineering Program (MCT)

MCT211	**Automatic Control**	**Spring 2025**

Project

Submitted by

Name	ID
Adham Hamada Hazem	22P0100
Abdullah Muhammad Fahiem	22P0034
Mahmoud Sami Mahmoud Yaghi	2200266
Shahd Ahmed Tawfik	23P0169
Rewan Mohamed Mostafa	22P0383
Ahmed Hisham	2200618
Ahmed Mohamed Behiry	22P0044

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1 Introduction

This project aims to design and implement a Proportional-Integral-Derivative (PID) controller to control the displacement of a cart in an electromechanical system. The project is divided into two phases:

Phase 1: PID controller for displacement control using the Ziegler-Nichols method.

Phase 2: Cascaded control system to control both the displacement and velocity of the cart using PID controllers.

2 System Modeling

2.1 Schematic presentation of the system:

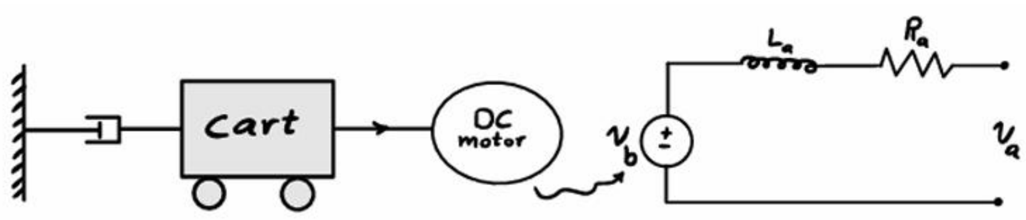


Figure 1: system schematic

2.2 System modeling in Simulink:

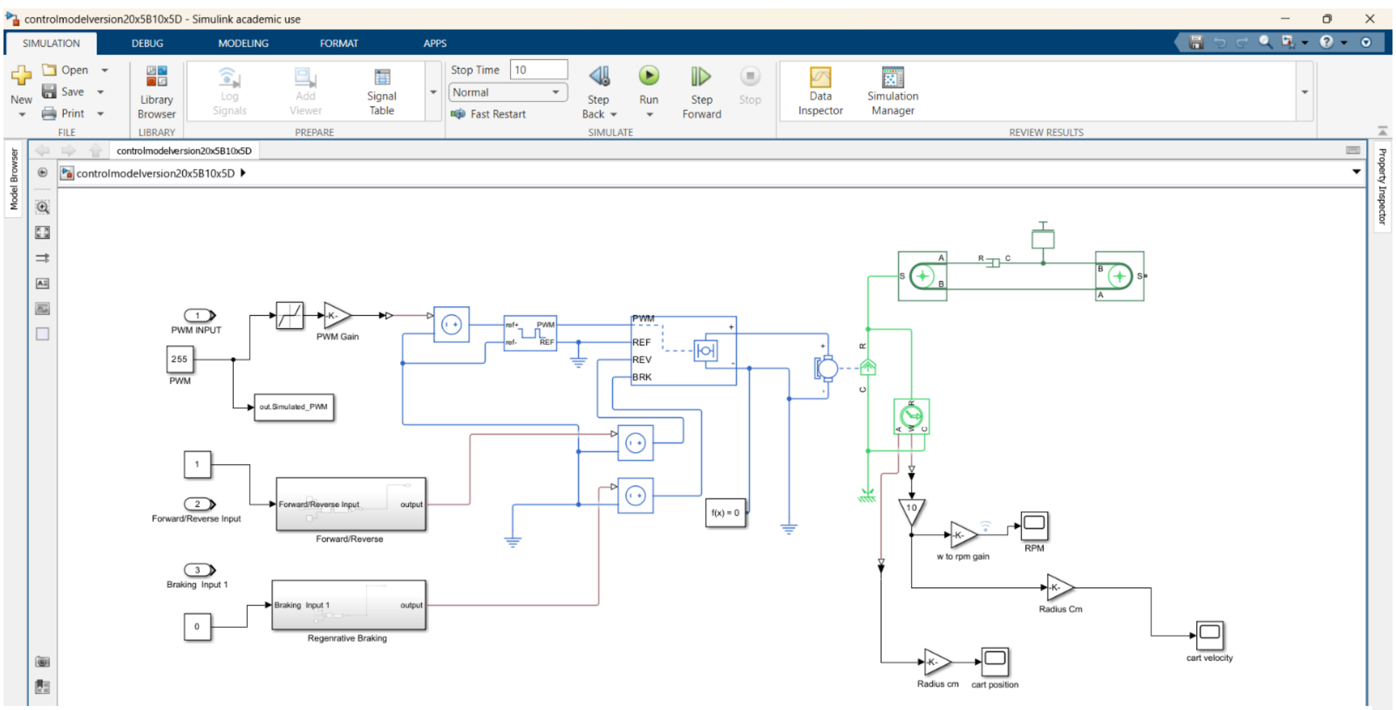


Figure 2: Simulink model

The presented Simulink model simulates the control logic for a DC motor-driven cart system. The system accepts user inputs to control the direction, speed (via PWM), and breaking the motor, which in turn drives the cart. The model incorporates motor control logic, direction control, regenerative breaking and kinematic calculation of velocity and position.

2.3 Parameter estimation:

To model correctly the parameters like motor parameters should be estimated by comparing response of simulation and physical model using parameter estimation in MATLAB.

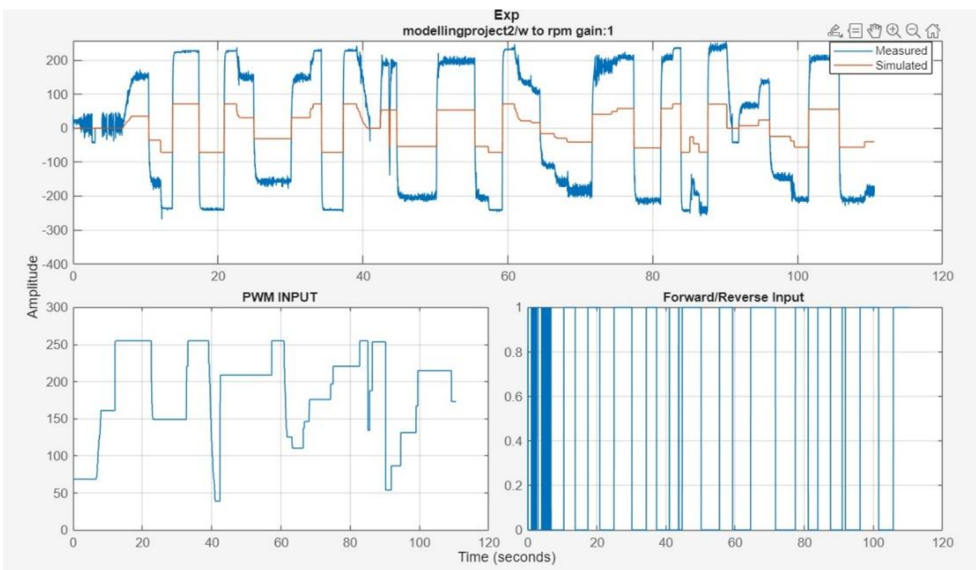


Figure 3: without parameter estimation

Applying low pass filter to reduce noise

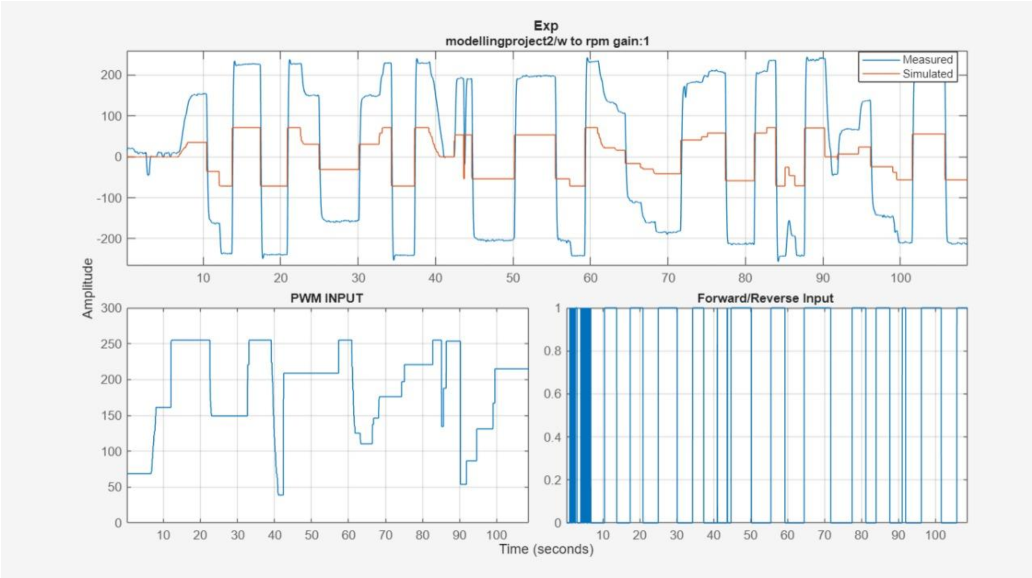


Figure 4: without parameter estimation with filter

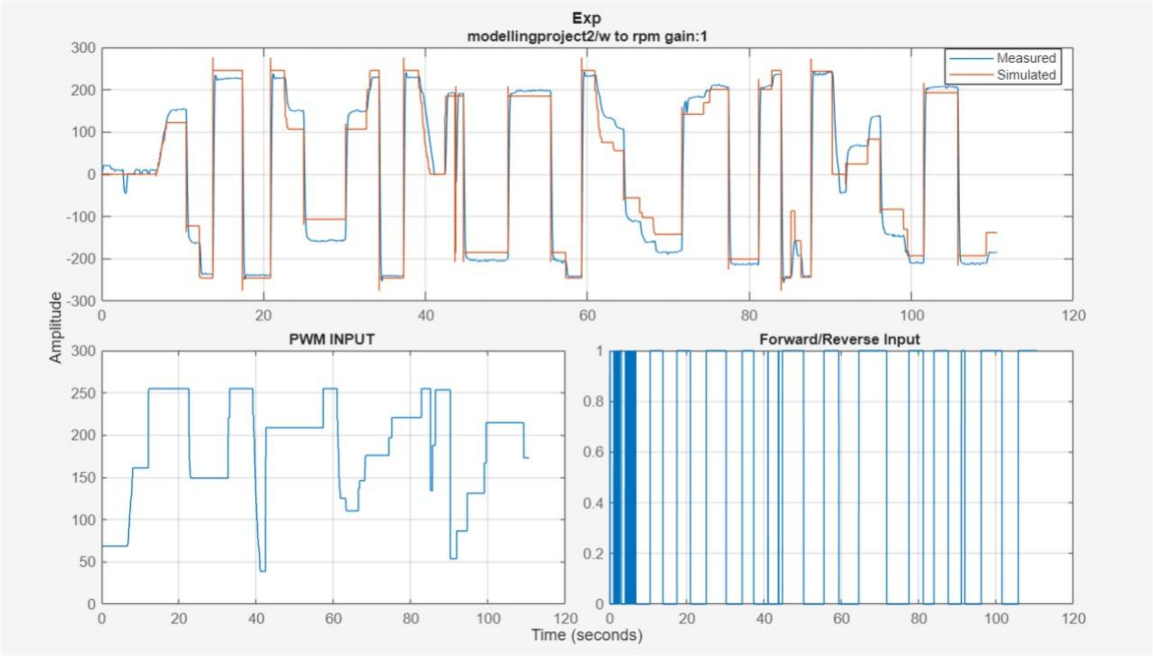


Figure5: After parameter estimation

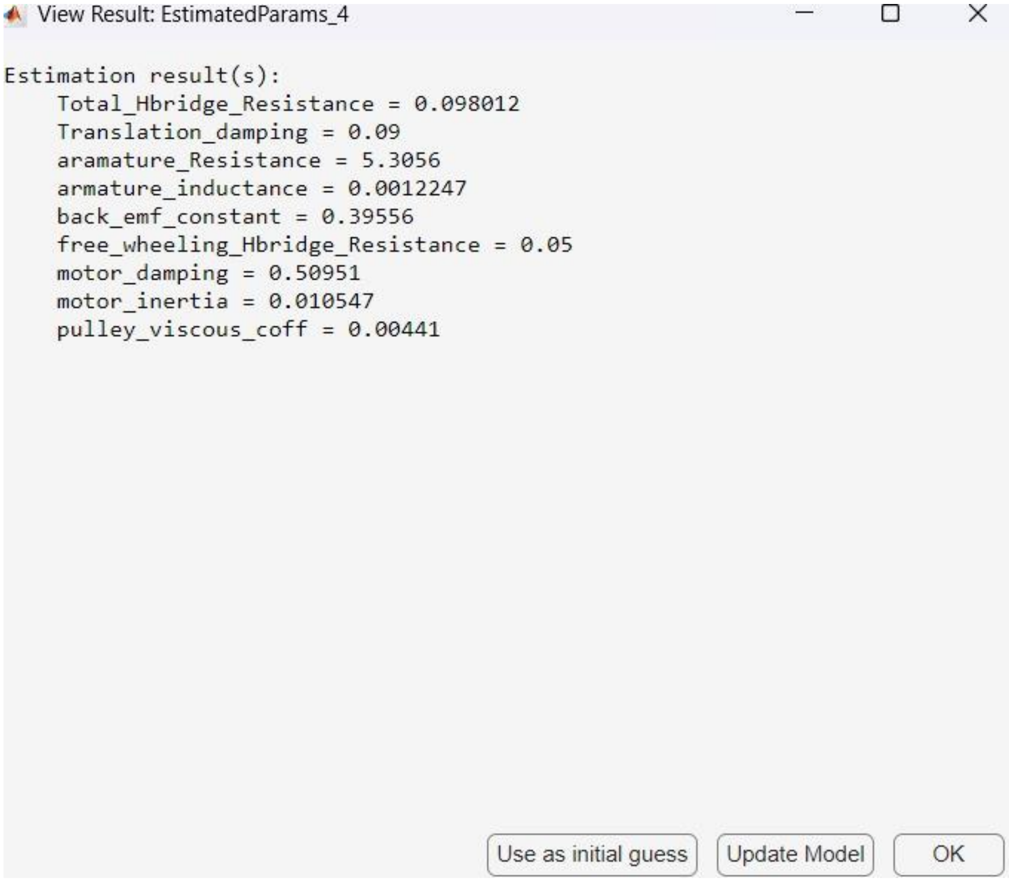
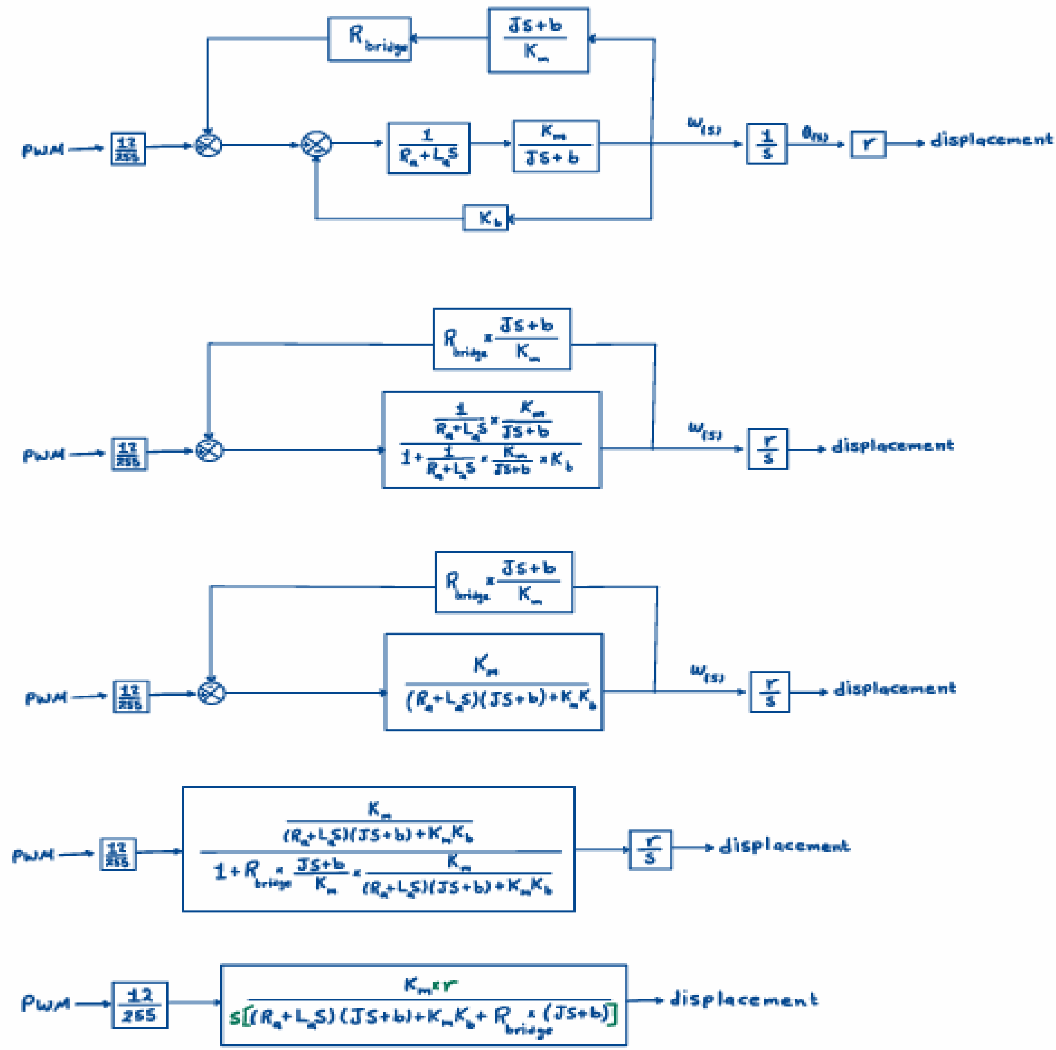


Figure 6: Estimated parameters

2.4 **Phase 1** Transfer function deduction:

Using parameters estimated and block diagram reduction transfer function between displacement and PWM can be deduced.



$b_{eq} = b_{motor} + b_{Pulley} + b_{cart} \times r^2$
$= 0.50951 + 0.00441 + 0.09(0.0075)^2 = 0.5139$

$J_{eq} = (m_{cart} + m_{Pulley}) r^2 + J$
$= (150\times10^{-3}) \times (0.0075)^2 + 0.010547 = 0.01055$

$\rightarrow (R_a+L_aS)(JS+b)$
$= L_aJS^2 + (R_aJ + L_ab)S + R_ab$
$= 1.292\times10^{-5} S^2 + 0.0566 S + 2.7265$

$\rightarrow R_bJS + R_bb$
$= 1.034\times10^{-3} S + 0.05036$

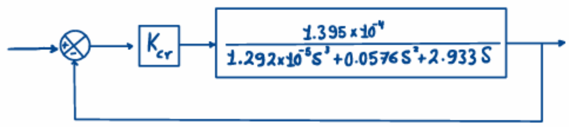
$\rightarrow K_b \simeq K_m,\ K_mK_b = K_b^2$
$= (0.039556)^2 = 0.15646$

$$\frac{displacement_{(S)}}{PWM_{(S)}} = \frac{\frac{12}{255} \times 0.39556 \times 0.0075}{1.292\times10^{-5}S^3 + 0.0576S^2 + 2.933S} = \frac{1.395\times10^{-4}}{1.292\times10^{-5}S^3 + 0.0576S^2 + 2.933S}$$

Figure 7:Displacement Transfer Function

3 PID Controller Design

3.1 Obtain closed loop function then using Routh to get value of K critical.



$$\frac{\dfrac{1.395\times10^{-4}K_{cr}}{1.292\times10^{-5}S^3+0.0576S^2+2.933S}}{1+\dfrac{1.395\times10^{-4}K_{cr}}{1.292\times10^{-5}S^3+0.0576S^2+2.933S}}=\frac{1.395\times10^{-4}K_{cr}}{1.292\times10^{-5}S^3+0.0576S^2+2.933S+1.395\times10^{-4}K_{cr}}$$

$$\begin{array}{lll}
s^3 & 1.291\times10^{-5} & 2.9345 \\
s^2 & 0.05764 & 1.395\times10^{-4}K_{cr} \\
s^1 & b_{n-1} & 0 \\
s^0 & 1.395\times10^{-4}K_{cr} &
\end{array}$$

$$b_{n-1}=\frac{-1}{0.05764}\begin{vmatrix}1.291\times10^{-5} & 2.9345 \\ 0.05764 & 1.395\times10^{-4}K_{cr}\end{vmatrix}=\frac{0.05764\times2.9345-1.291\times10^{-5}\times(1.395\times10^{-4}K_{cr})}{0.05764}=0\rightarrow K_{cr}=9.395\times10^{7}$$

Designing a PID controller derives the transfer function that represents the controller's behavior. For a standard PID controller, the transfer function is given by:

$$G_{pID}(s)=k_p+\frac{k_i}{s}+k_d s$$

Where:

- K_p is the proportional gain.
- K_i is the integral gain.
- K_d is the derivative gain.
- s is the Laplace variable (complex frequency).

Then, we calculated the initial values for Kp, Ki and Kd using Ziegler Nicholas method. However, the response was not satisfactory. So, we opted to manual tuning.

3.2 Tuning to get the controller parameters (Kp, Ki and Kd):

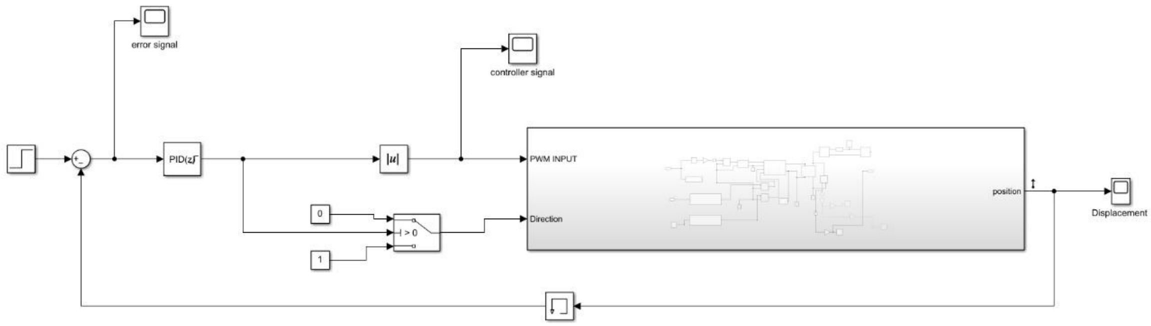


Figure 8: Controller on Subsystem Model

As stated, we decided that manual tuning would be our approach. After some iterations, we found that a PD controller was the best fit, with Kp = 1000 and Kd = 0.5.

Figure 9: Displacement Response to 40 cm Setpoint

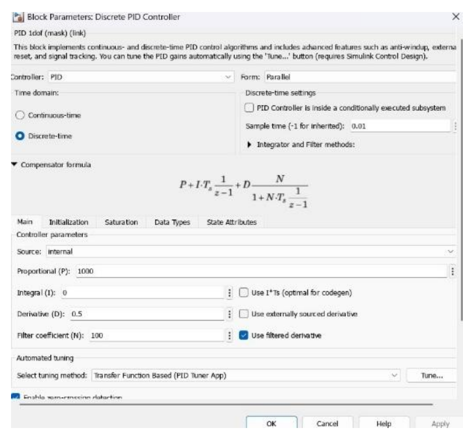


Figure 10: Kp and Kd values

Figure 11: Error Signal

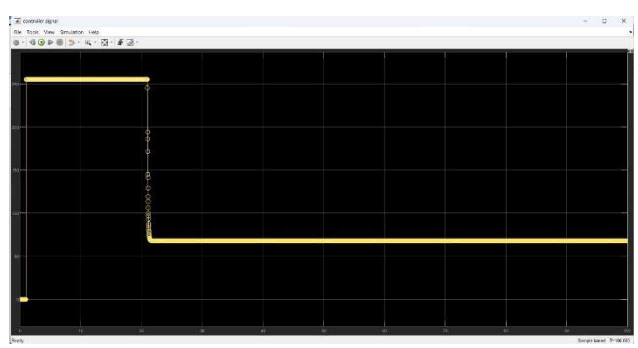
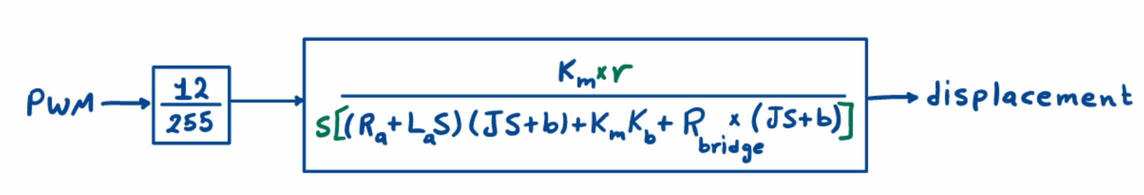


Figure 12: Controller Signal

4 Cascaded Control Design

The cascaded control system is designed to regulate the displacement and velocity of a cart by organizing two PID control loops in a hierarchical structure. The **outer loop** controls the cart's **displacement (position)**, while the **inner loop** regulates its **velocity**. This design improves system performance by allowing the inner loop to quickly respond to dynamic changes, stabilizing the system before the outer loop acts.



By using transfer function of displacement, we can obtain transfer function between **angular velocity(W)** and **PWM** by removing r/s term. To convert it to **RPM** and **PWM** transfer function we add gain of 60/2π.

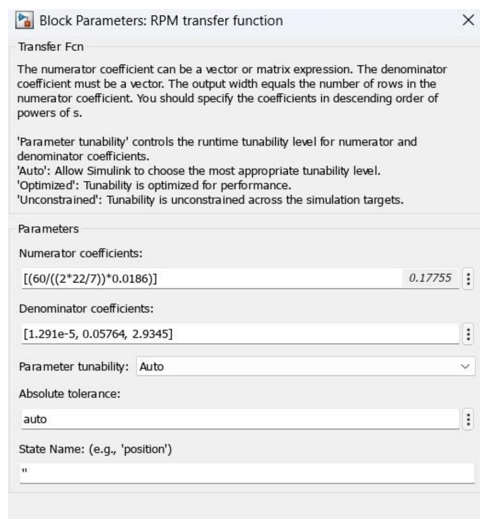


Figure 13: Transfer function velocity

Tuning Process:

1. **Tune the inner velocity loop first**, using step inputs and adjusting PID parameters to achieve a fast and stable response with minimal overshoot.
2. **Once the inner loop is stable**, tune the outer displacement loop. The velocity loop's fast response helps isolate and simplify the tuning of the outer loop.
3. Both controllers are fine-tuned through simulation in MATLAB/Simulink and validated with physical system tests, using encoder feedback for accurate measurement.

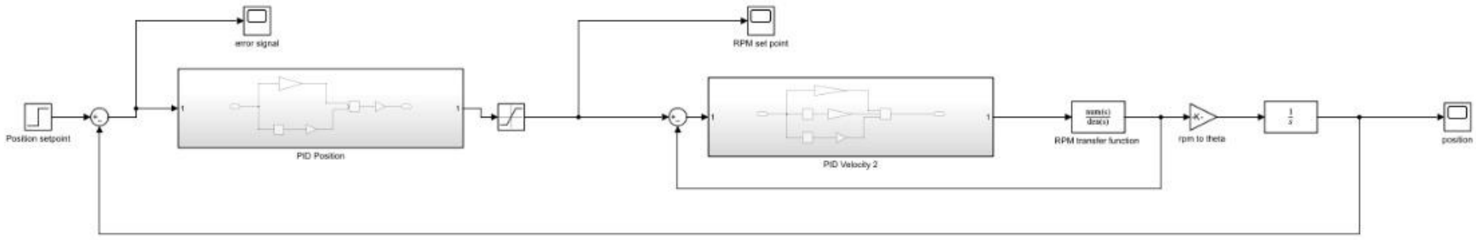


Figure 14: Cascaded PID controller

Tuning Results:

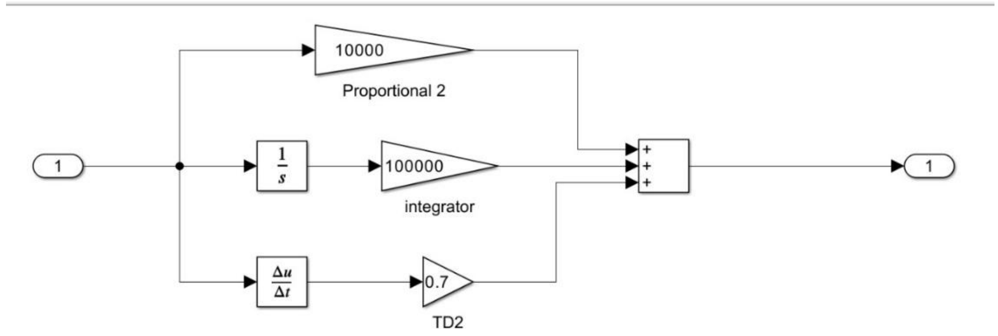


Figure 15: PID inner loop parameters

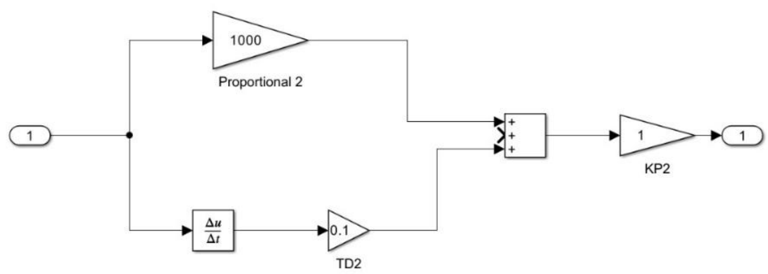


Figure 16:PID outer loop parameters

Response of the system:

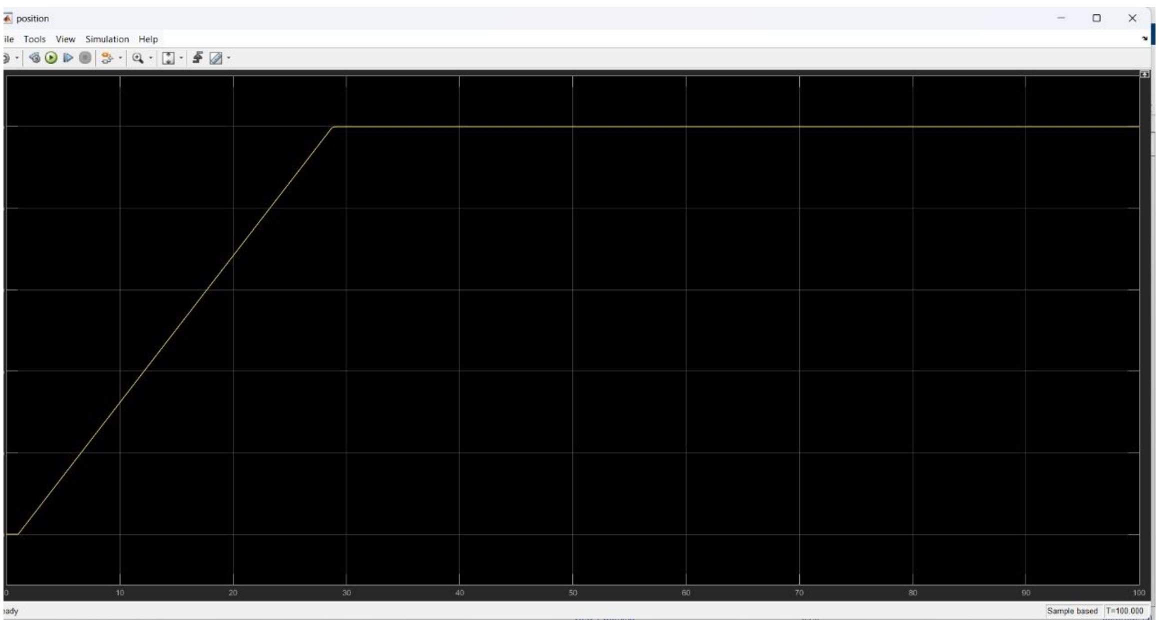


Figure 17: cascaded response at set point 50

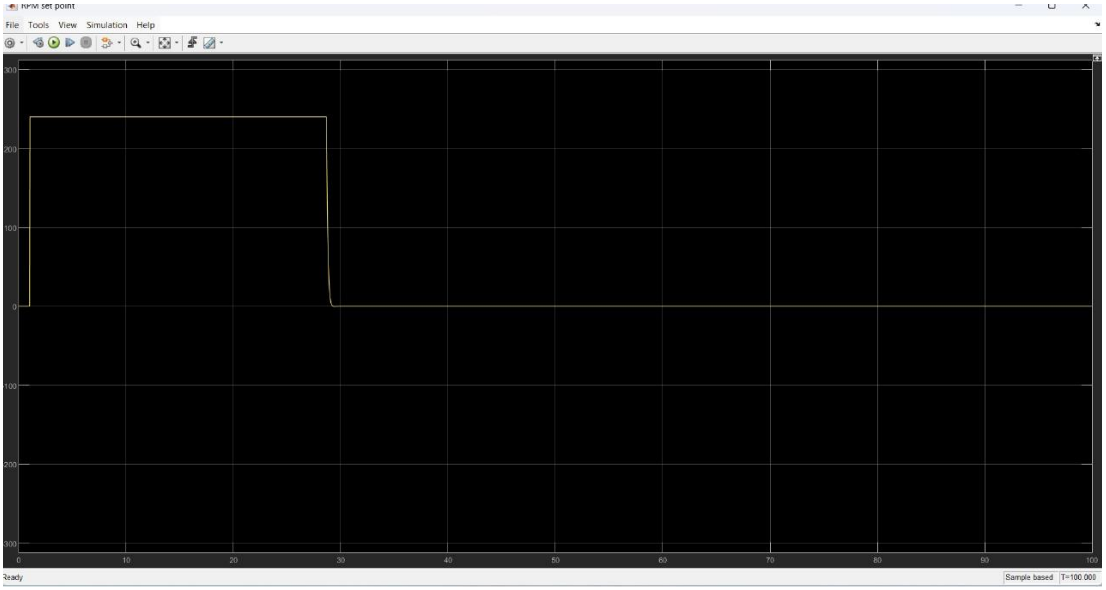


Figure 18: RPM set point

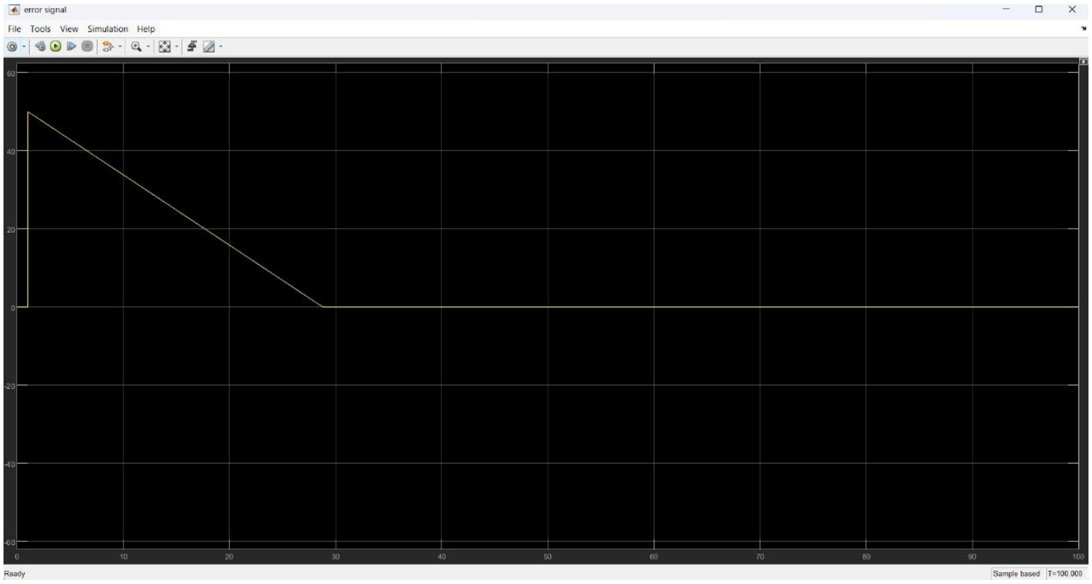
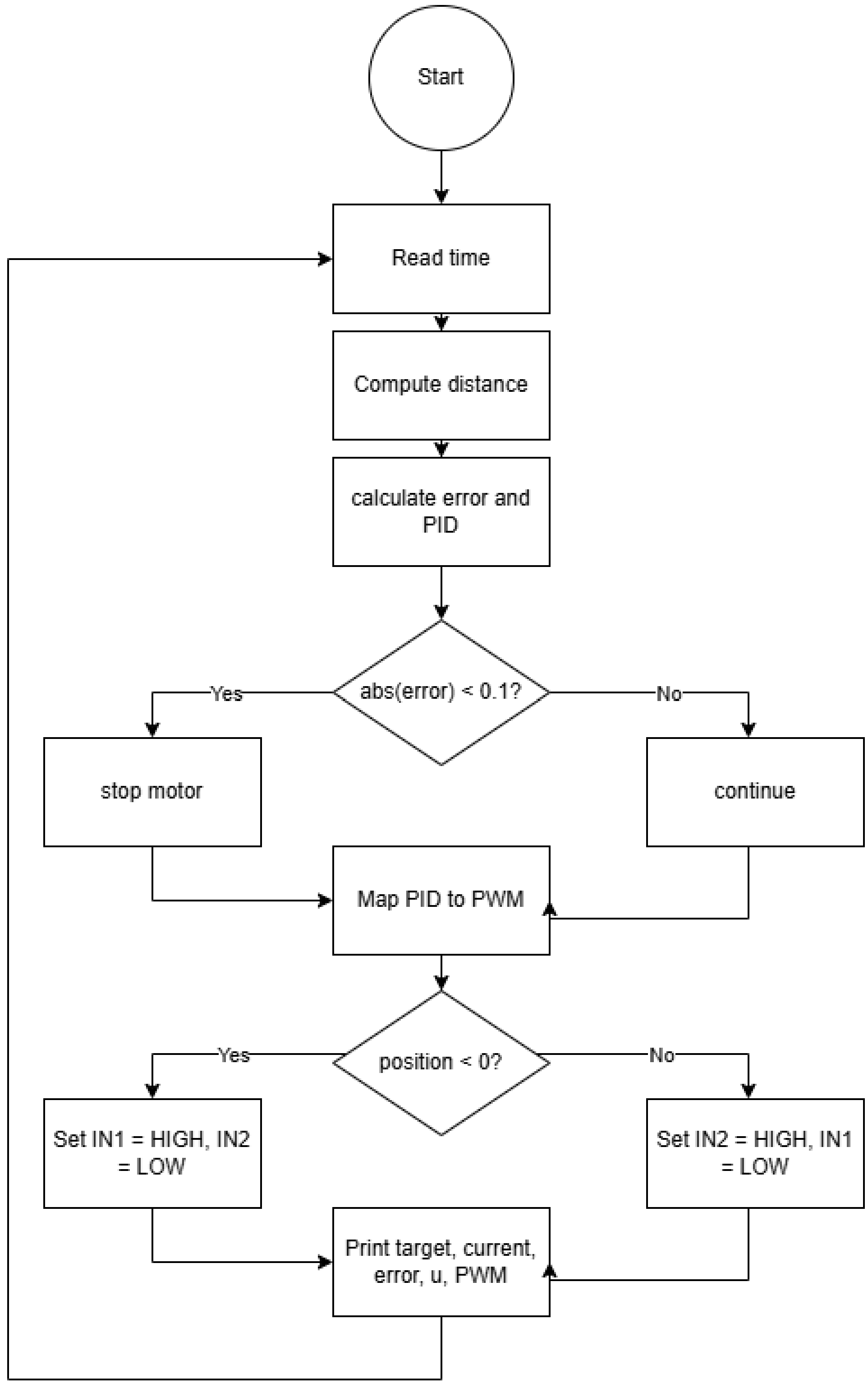


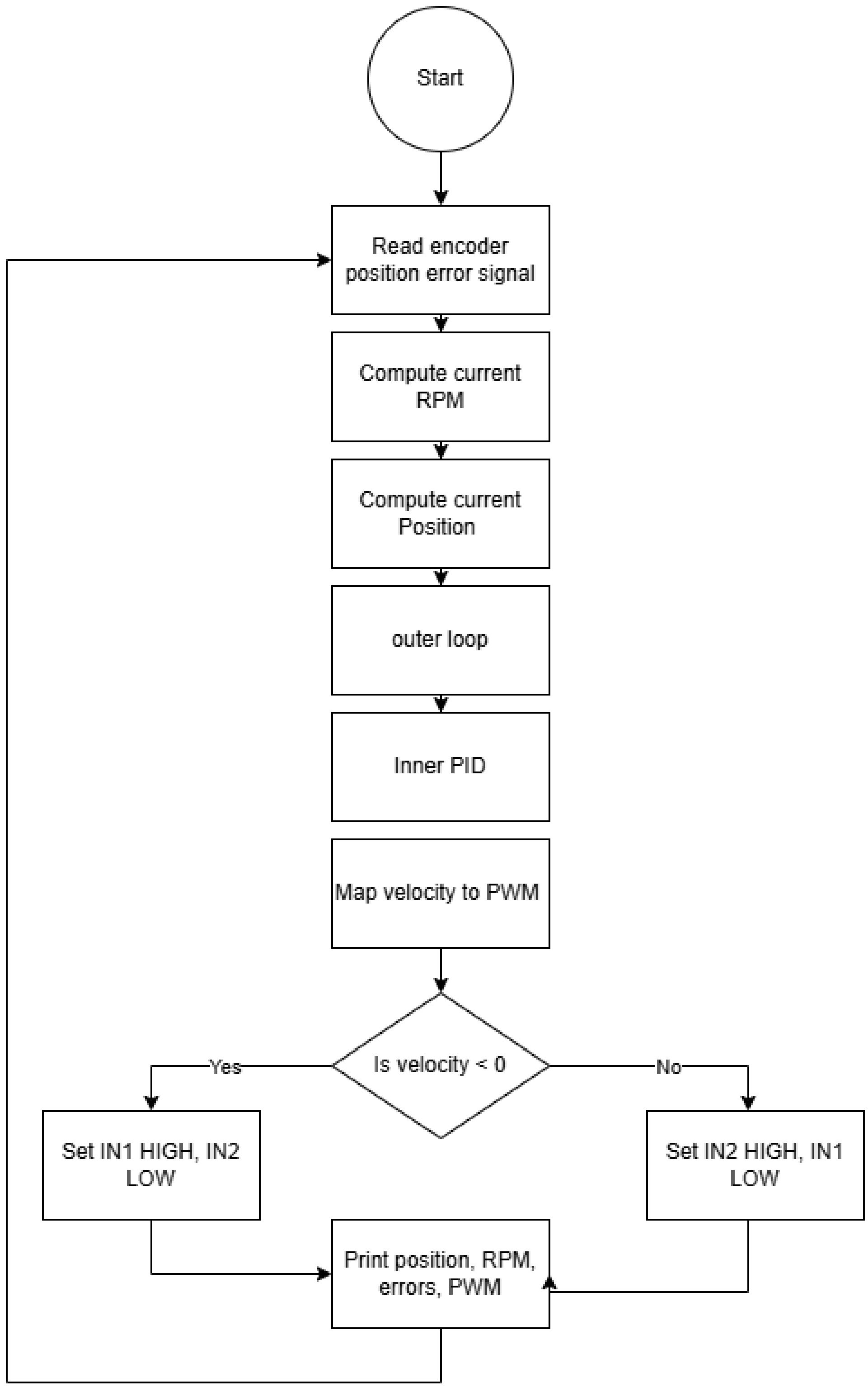
Figure 19: error signal

5 Code flow chart

5.1 Displacement Flow chart



5.2 Cascaded control



6 Simulation and code files

https://drive.google.com/drive/folders/1oZnErltjdHJS-HzABkFlfKB4gpOpf5kP?usp=sharing